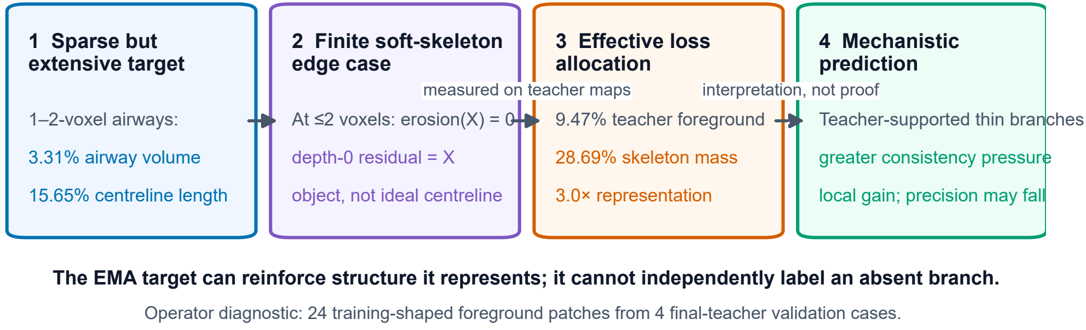
1 Sparse but extensive target
1–2-voxel airways:
3.31% airway volume
15.65% centreline length
2 Finite soft-skeleton edge case
At ≤2 voxels: erosion(X) = 0
depth-0 residual = X
object, not ideal centreline
measured on teacher maps
3 Effective loss allocation
9.47% teacher foreground
28.69% skeleton mass
3.0× representation
interpretation, not proof
4 Mechanistic prediction
Teacher-supported thin branches
greater consistency pressure
local gain; precision may fall
The EMA target can reinforce structure it represents; it cannot independently label an absent branch.
Operator diagnostic: 24 training-shaped foreground patches from 4 final-teacher validation cases.